



TALENT MATCH AI

AI-Based Job-Resume Matching System

INTRODUCTION

- Talent Match AI is an AI-powered system that automates the process of matching job descriptions with candidate resumes. Unlike traditional keyword-based methods, it uses advanced Natural Language Processing and models like BERT to understand the semantic meaning of text. This enables accurate classification and ranking of candidates based on relevance, reducing manual effort and improving hiring efficiency.

PROBLEM STATEMENT

- Manual resume screening is time-consuming and inefficient
- Keyword-based matching fails to capture real meaning
- Qualified candidates may be missed due to poor matching

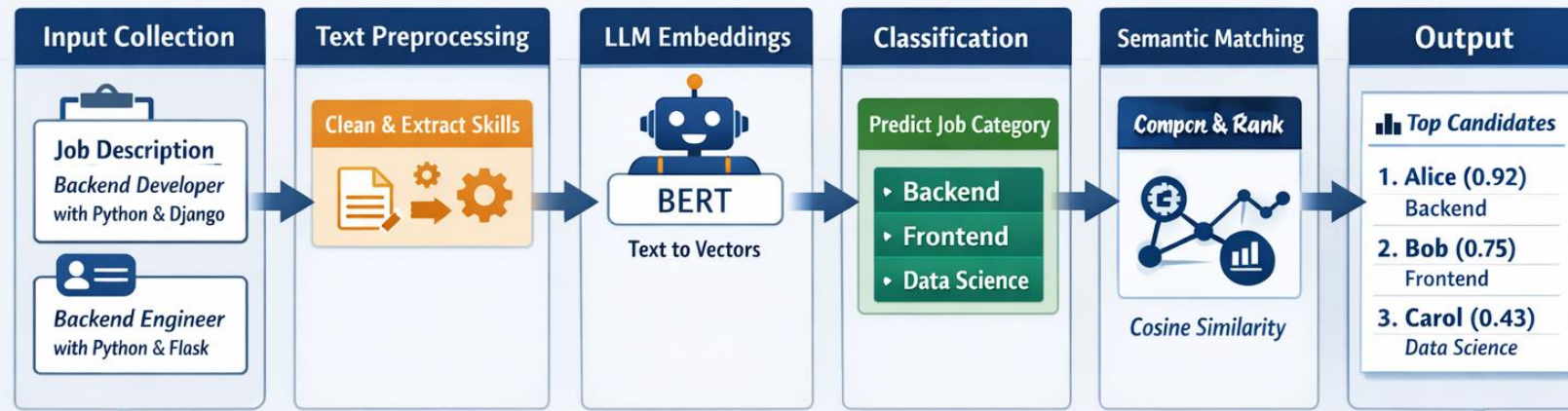
OBJECTIVES

- Automate job–resume matching using AI
- Classify resumes and job roles accurately
- Rank candidates based on semantic similarity
- Reduce HR workload and improve hiring efficiency

SYSTEM WORKFLOW OVERVIEW

- Input Collection - Collect job descriptions
- Preprocessing - Clean structured text
- Embeddings - Use BERT model
- Classification - Categorize into job roles
- Matching - Compare job vs resume vectors
- Output - Job category prediction

TalentMatchAI Workflow



AI-Powered Matching



Semantic Analysis



Confidence Scores

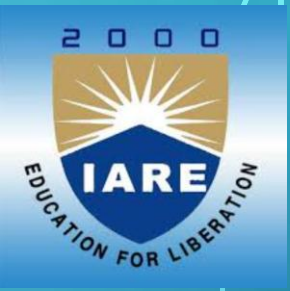
TECHNOLOGIES USED

- Python, Pandas
- Transformers (BERT)
- Scikit-learn
- Stream lit

ADVANTAGES

- Understands semantic meaning, not just keywords
- Improves accuracy in candidate-job matching
- Reduces manual effort for HR teams
- Scalable for large numbers of resumes and jobs

OUTPUT



► Interactive Demo

Try TalentMatchAI Live

Paste a resume below and our AI will classify it, extract skills, and rank it against sample job descriptions in real-time.



This demo uses an LLM to simulate the classification and matching pipeline described in the project guide. In production, you would use BERT embeddings and a fine-tuned classifier for faster, more consistent results.

✦ Input Resume

📄 Resume Text

Paste a resume or candidate profile here...

Or try a sample:

Backend Developer

Frontend Developer

Data Scientist

Analysis Results



Enter a resume and click "Analyze & Match"
Results will appear here

<https://spiritual-talent-match-flow.base44.app>

CONCLUSION

- Talent Match AI provides an intelligent and efficient solution for automating the recruitment process by using AI to understand the semantic meaning of resumes and job descriptions. It improves the accuracy of candidate matching, reduces manual effort, and enables faster hiring decisions. Overall, the system enhances recruitment quality and represents a scalable approach for modern hiring needs.

The background is a blue gradient. In the corners, there are white line-art illustrations of circuit boards or neural networks, with lines and small circles representing nodes.

THANK YOU